Material suplementario

Réplica y validación del controlador de lógica difusa para una intersección de tráfico de Pappis y Mamdani (1977): un estudio de reproducibilidad

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Este documento acompaña al artículo principal. Las referencias a secciones, ecuaciones y tablas del artículo principal usan su numeración (las tablas y figuras de este suplemento se numeran S1, S2, ...). Contiene: (S1) la lista literal de las 180 reglas del espacio $T \times A \times Q$ con las 25 reglas del original resaltadas; (S2) los seis apartados del simulador por separado; (S3) figuras adicionales; y (S4) tablas adicionales.

1. Las 180 reglas del espacio $T \times A \times Q$

Cada regla tiene la forma «si $T = t$ y $A = mt(a)$ y $Q = lt(q)$ entonces $E = t$». Las 180 reglas son las $5 \times 6 \times 6$ combinaciones de las etiquetas de las Tablas II a IV del original; el original usa 25 de ellas (cinco por intervención), resaltadas en verde y marcadas con ✓; las 161 restantes se marcan con ×. Las 25 reglas ocupan 19 filas porque varias se repiten entre intervenciones y la regla R1 usa $Q =$ any (marcada R1*), que cubre las seis filas de $T =$ very short y $A =$ none. La matriz compacta está en el Apéndice A del artículo principal; los mismos datos están en `resultados/reglas_180.csv`.

Tabla S1: Las 180 reglas del espacio $T \times A \times Q$ ($5 \times 6 \times 6$; cada una concluye $E = T$). Las que usa el autor están resaltadas en verde y marcadas con ✓; las demás, con ×. Las columnas de intervención indican con qué número de regla (R1 a R5) la usa cada intervención; R1* es la regla R1 del autor, que usa $Q =$ any (no condiciona la cola) y por eso cubre las seis reglas de su fila.

#	Regla	7 s	17 s	27 s	37 s	47 s	Usada
1	**SI** T = very short **Y** A = mt(none) **Y** Q = lt(very small) **ENTONCES** E = very short	R1*	R1*	R1*	R1*	R1*	✓
2	**SI** T = very short **Y** A = mt(none) **Y** Q = lt(small) **ENTONCES** E = very short	R1*	R1*	R1*	R1*	R1*	✓
3	**SI** T = very short **Y** A = mt(none) **Y** Q = lt(small plus) **ENTONCES** E = very short	R1*	R1*	R1*	R1*	R1*	✓
4	**SI** T = very short **Y** A = mt(none) **Y** Q = lt(medium) **ENTONCES** E = very short	R1*	R1*	R1*	R1*	R1*	✓
5	**SI** T = very short **Y** A = mt(none) **Y** Q = lt(long) **ENTONCES** E = very short	R1*	R1*	R1*	R1*	R1*	✓
6	**SI** T = very short **Y** A = mt(none) **Y** Q = lt(very long) **ENTONCES** E = very short	R1*	R1*	R1*	R1*	R1*	✓
7	**SI** T = very short **Y** A = mt(a few) **Y** Q = lt(very small) **ENTONCES** E = very short						×
8	**SI** T = very short **Y** A = mt(a few) **Y** Q = lt(small) **ENTONCES** E = very short						×
9	**SI** T = very short **Y** A = mt(a few) **Y** Q = lt(small plus) **ENTONCES** E = very short						×
10	**SI** T = very short **Y** A = mt(a few) **Y** Q = lt(medium) **ENTONCES** E = very short						×
11	**SI** T = very short **Y** A = mt(a few) **Y** Q = lt(long) **ENTONCES** E = very short						×
12	**SI** T = very short **Y** A = mt(a few) **Y** Q = lt(very long) **ENTONCES** E = very short						×
13	**SI** T = very short **Y** A = mt(few) **Y** Q = lt(very small) **ENTONCES** E = very short						×
14	**SI** T = very short **Y** A = mt(few) **Y** Q = lt(small) **ENTONCES** E = very short						×
15	**SI** T = very short **Y** A = mt(few) **Y** Q = lt(small plus) **ENTONCES** E = very short						×
16	**SI** T = very short **Y** A = mt(few) **Y** Q = lt(medium) **ENTONCES** E = very short						×
17	**SI** T = very short **Y** A = mt(few) **Y** Q = lt(long) **ENTONCES** E = very short						×
18	**SI** T = very short **Y** A = mt(few) **Y** Q = lt(very long) **ENTONCES** E = very short						×
19	**SI** T = very short **Y** A = mt(medium) **Y** Q = lt(very small) **ENTONCES** E = very short						×
20	**SI** T = very short **Y** A = mt(medium) **Y** Q = lt(small) **ENTONCES** E = very short						×
21	**SI** T = very short **Y** A = mt(medium) **Y** Q = lt(small plus) **ENTONCES** E = very short						×
22	**SI** T = very short **Y** A = mt(medium) **Y** Q = lt(medium) **ENTONCES** E = very short						×
23	**SI** T = very short **Y** A = mt(medium) **Y** Q = lt(long) **ENTONCES** E = very short						×
24	**SI** T = very short **Y** A = mt(medium) **Y** Q = lt(very long) **ENTONCES** E = very short						×
25	**SI** T = very short **Y** A = mt(many) **Y** Q = lt(very small) **ENTONCES** E = very short						×
26	**SI** T = very short **Y** A = mt(many) **Y** Q = lt(small) **ENTONCES** E = very short						×
27	**SI** T = very short **Y** A = mt(many) **Y** Q = lt(small plus) **ENTONCES** E = very short						×
28	**SI** T = very short **Y** A = mt(many) **Y** Q = lt(medium) **ENTONCES** E = very short						×
29	**SI** T = very short **Y** A = mt(many) **Y** Q = lt(long) **ENTONCES** E = very short						×
30	**SI** T = very short **Y** A = mt(many) **Y** Q = lt(very long) **ENTONCES** E = very short						×
31	**SI** T = very short **Y** A = mt(too many) **Y** Q = lt(very small) **ENTONCES** E = very short						×
32	**SI** T = very short **Y** A = mt(too many) **Y** Q = lt(small) **ENTONCES** E = very short						×
33	**SI** T = very short **Y** A = mt(too many) **Y** Q = lt(small plus) **ENTONCES** E = very short						×
34	**SI** T = very short **Y** A = mt(too many) **Y** Q = lt(medium) **ENTONCES** E = very short						×
35	**SI** T = very short **Y** A = mt(too many) **Y** Q = lt(long) **ENTONCES** E = very short						×
36	**SI** T = very short **Y** A = mt(too many) **Y** Q = lt(very long) **ENTONCES** E = very short						×
37	**SI** T = short **Y** A = mt(none) **Y** Q = lt(very small) **ENTONCES** E = short						×
38	**SI** T = short **Y** A = mt(none) **Y** Q = lt(small) **ENTONCES** E = short						×
39	**SI** T = short **Y** A = mt(none) **Y** Q = lt(small plus) **ENTONCES** E = short						×
40	**SI** T = short **Y** A = mt(none) **Y** Q = lt(medium) **ENTONCES** E = short						×
41	**SI** T = short **Y** A = mt(none) **Y** Q = lt(long) **ENTONCES** E = short						×
42	**SI** T = short **Y** A = mt(none) **Y** Q = lt(very long) **ENTONCES** E = short						×
43	**SI** T = short **Y** A = mt(a few) **Y** Q = lt(very small) **ENTONCES** E = short	R2	R2	R2			✓
44	**SI** T = short **Y** A = mt(a few) **Y** Q = lt(small) **ENTONCES** E = short						×
45	**SI** T = short **Y** A = mt(a few) **Y** Q = lt(small plus) **ENTONCES** E = short				R2		✓
46	**SI** T = short **Y** A = mt(a few) **Y** Q = lt(medium) **ENTONCES** E = short						×
47	**SI** T = short **Y** A = mt(a few) **Y** Q = lt(long) **ENTONCES** E = short					R2	✓
48	**SI** T = short **Y** A = mt(a few) **Y** Q = lt(very long) **ENTONCES** E = short						×
49	**SI** T = short **Y** A = mt(few) **Y** Q = lt(very small) **ENTONCES** E = short						×
50	**SI** T = short **Y** A = mt(few) **Y** Q = lt(small) **ENTONCES** E = short						×

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#	Regla	7 s	17 s	27 s	37 s	47 s	Usada
51	**SI** T = short **Y** A = mt(few) **Y** Q = lt(small plus) **ENTONCES** E = short						×
52	**SI** T = short **Y** A = mt(few) **Y** Q = lt(medium) **ENTONCES** E = short						×
53	**SI** T = short **Y** A = mt(few) **Y** Q = lt(long) **ENTONCES** E = short						×
54	**SI** T = short **Y** A = mt(few) **Y** Q = lt(very long) **ENTONCES** E = short						×
55	**SI** T = short **Y** A = mt(medium) **Y** Q = lt(very small) **ENTONCES** E = short						×
56	**SI** T = short **Y** A = mt(medium) **Y** Q = lt(small) **ENTONCES** E = short						×
57	**SI** T = short **Y** A = mt(medium) **Y** Q = lt(small plus) **ENTONCES** E = short						×
58	**SI** T = short **Y** A = mt(medium) **Y** Q = lt(medium) **ENTONCES** E = short						×
59	**SI** T = short **Y** A = mt(medium) **Y** Q = lt(long) **ENTONCES** E = short						×
60	**SI** T = short **Y** A = mt(medium) **Y** Q = lt(very long) **ENTONCES** E = short						×
61	**SI** T = short **Y** A = mt(many) **Y** Q = lt(very small) **ENTONCES** E = short						×
62	**SI** T = short **Y** A = mt(many) **Y** Q = lt(small) **ENTONCES** E = short						×
63	**SI** T = short **Y** A = mt(many) **Y** Q = lt(small plus) **ENTONCES** E = short						×
64	**SI** T = short **Y** A = mt(many) **Y** Q = lt(medium) **ENTONCES** E = short						×
65	**SI** T = short **Y** A = mt(many) **Y** Q = lt(long) **ENTONCES** E = short						×
66	**SI** T = short **Y** A = mt(many) **Y** Q = lt(very long) **ENTONCES** E = short						×
67	**SI** T = short **Y** A = mt(too many) **Y** Q = lt(very small) **ENTONCES** E = short						×
68	**SI** T = short **Y** A = mt(too many) **Y** Q = lt(small) **ENTONCES** E = short						×
69	**SI** T = short **Y** A = mt(too many) **Y** Q = lt(small plus) **ENTONCES** E = short						×
70	**SI** T = short **Y** A = mt(too many) **Y** Q = lt(medium) **ENTONCES** E = short						×
71	**SI** T = short **Y** A = mt(too many) **Y** Q = lt(long) **ENTONCES** E = short						×
72	**SI** T = short **Y** A = mt(too many) **Y** Q = lt(very long) **ENTONCES** E = short						×
73	**SI** T = medium **Y** A = mt(none) **Y** Q = lt(very small) **ENTONCES** E = medium						×
74	**SI** T = medium **Y** A = mt(none) **Y** Q = lt(small) **ENTONCES** E = medium						×
75	**SI** T = medium **Y** A = mt(none) **Y** Q = lt(small plus) **ENTONCES** E = medium						×
76	**SI** T = medium **Y** A = mt(none) **Y** Q = lt(medium) **ENTONCES** E = medium						×
77	**SI** T = medium **Y** A = mt(none) **Y** Q = lt(long) **ENTONCES** E = medium						×
78	**SI** T = medium **Y** A = mt(none) **Y** Q = lt(very long) **ENTONCES** E = medium						×
79	**SI** T = medium **Y** A = mt(a few) **Y** Q = lt(very small) **ENTONCES** E = medium						×
80	**SI** T = medium **Y** A = mt(a few) **Y** Q = lt(small) **ENTONCES** E = medium						×
81	**SI** T = medium **Y** A = mt(a few) **Y** Q = lt(small plus) **ENTONCES** E = medium						×
82	**SI** T = medium **Y** A = mt(a few) **Y** Q = lt(medium) **ENTONCES** E = medium						×
83	**SI** T = medium **Y** A = mt(a few) **Y** Q = lt(long) **ENTONCES** E = medium						×
84	**SI** T = medium **Y** A = mt(a few) **Y** Q = lt(very long) **ENTONCES** E = medium						×
85	**SI** T = medium **Y** A = mt(few) **Y** Q = lt(very small) **ENTONCES** E = medium	R3	R3	R3			✓
86	**SI** T = medium **Y** A = mt(few) **Y** Q = lt(small) **ENTONCES** E = medium						×
87	**SI** T = medium **Y** A = mt(few) **Y** Q = lt(small plus) **ENTONCES** E = medium						×
88	**SI** T = medium **Y** A = mt(few) **Y** Q = lt(medium) **ENTONCES** E = medium						×
89	**SI** T = medium **Y** A = mt(few) **Y** Q = lt(long) **ENTONCES** E = medium						×
90	**SI** T = medium **Y** A = mt(few) **Y** Q = lt(very long) **ENTONCES** E = medium						×
91	**SI** T = medium **Y** A = mt(medium) **Y** Q = lt(very small) **ENTONCES** E = medium						×
92	**SI** T = medium **Y** A = mt(medium) **Y** Q = lt(small) **ENTONCES** E = medium						×
93	**SI** T = medium **Y** A = mt(medium) **Y** Q = lt(small plus) **ENTONCES** E = medium				R3		✓
94	**SI** T = medium **Y** A = mt(medium) **Y** Q = lt(medium) **ENTONCES** E = medium						×
95	**SI** T = medium **Y** A = mt(medium) **Y** Q = lt(long) **ENTONCES** E = medium					R3	✓
96	**SI** T = medium **Y** A = mt(medium) **Y** Q = lt(very long) **ENTONCES** E = medium						×
97	**SI** T = medium **Y** A = mt(many) **Y** Q = lt(very small) **ENTONCES** E = medium						×
98	**SI** T = medium **Y** A = mt(many) **Y** Q = lt(small) **ENTONCES** E = medium						×
99	**SI** T = medium **Y** A = mt(many) **Y** Q = lt(small plus) **ENTONCES** E = medium						×
100	**SI** T = medium **Y** A = mt(many) **Y** Q = lt(medium) **ENTONCES** E = medium						×
101	**SI** T = medium **Y** A = mt(many) **Y** Q = lt(long) **ENTONCES** E = medium						×
102	**SI** T = medium **Y** A = mt(many) **Y** Q = lt(very long) **ENTONCES** E = medium						×
103	**SI** T = medium **Y** A = mt(too many) **Y** Q = lt(very small) **ENTONCES** E = medium						×
104	**SI** T = medium **Y** A = mt(too many) **Y** Q = lt(small) **ENTONCES** E = medium						×
105	**SI** T = medium **Y** A = mt(too many) **Y** Q = lt(small plus) **ENTONCES** E = medium						×
106	**SI** T = medium **Y** A = mt(too many) **Y** Q = lt(medium) **ENTONCES** E = medium						×
107	**SI** T = medium **Y** A = mt(too many) **Y** Q = lt(long) **ENTONCES** E = medium						×
108	**SI** T = medium **Y** A = mt(too many) **Y** Q = lt(very long) **ENTONCES** E = medium						×
109	**SI** T = long **Y** A = mt(none) **Y** Q = lt(very small) **ENTONCES** E = long						×
110	**SI** T = long **Y** A = mt(none) **Y** Q = lt(small) **ENTONCES** E = long						×
111	**SI** T = long **Y** A = mt(none) **Y** Q = lt(small plus) **ENTONCES** E = long						×
112	**SI** T = long **Y** A = mt(none) **Y** Q = lt(medium) **ENTONCES** E = long						×
113	**SI** T = long **Y** A = mt(none) **Y** Q = lt(long) **ENTONCES** E = long						×
114	**SI** T = long **Y** A = mt(none) **Y** Q = lt(very long) **ENTONCES** E = long						×
115	**SI** T = long **Y** A = mt(a few) **Y** Q = lt(very small) **ENTONCES** E = long						×
116	**SI** T = long **Y** A = mt(a few) **Y** Q = lt(small) **ENTONCES** E = long						×
117	**SI** T = long **Y** A = mt(a few) **Y** Q = lt(small plus) **ENTONCES** E = long						×
118	**SI** T = long **Y** A = mt(a few) **Y** Q = lt(medium) **ENTONCES** E = long						×
119	**SI** T = long **Y** A = mt(a few) **Y** Q = lt(long) **ENTONCES** E = long						×
120	**SI** T = long **Y** A = mt(a few) **Y** Q = lt(very long) **ENTONCES** E = long						×
121	**SI** T = long **Y** A = mt(few) **Y** Q = lt(very small) **ENTONCES** E = long						×
122	**SI** T = long **Y** A = mt(few) **Y** Q = lt(small) **ENTONCES** E = long						×
123	**SI** T = long **Y** A = mt(few) **Y** Q = lt(small plus) **ENTONCES** E = long						×
124	**SI** T = long **Y** A = mt(few) **Y** Q = lt(medium) **ENTONCES** E = long						×
125	**SI** T = long **Y** A = mt(few) **Y** Q = lt(long) **ENTONCES** E = long						×
126	**SI** T = long **Y** A = mt(few) **Y** Q = lt(very long) **ENTONCES** E = long						×
127	**SI** T = long **Y** A = mt(medium) **Y** Q = lt(very small) **ENTONCES** E = long	R4	R4	R4			✓
128	**SI** T = long **Y** A = mt(medium) **Y** Q = lt(small) **ENTONCES** E = long						×
129	**SI** T = long **Y** A = mt(medium) **Y** Q = lt(small plus) **ENTONCES** E = long						×
130	**SI** T = long **Y** A = mt(medium) **Y** Q = lt(medium) **ENTONCES** E = long						×
131	**SI** T = long **Y** A = mt(medium) **Y** Q = lt(long) **ENTONCES** E = long						×
132	**SI** T = long **Y** A = mt(medium) **Y** Q = lt(very long) **ENTONCES** E = long						×
133	**SI** T = long **Y** A = mt(many) **Y** Q = lt(very small) **ENTONCES** E = long						×
134	**SI** T = long **Y** A = mt(many) **Y** Q = lt(small) **ENTONCES** E = long						×
135	**SI** T = long **Y** A = mt(many) **Y** Q = lt(small plus) **ENTONCES** E = long						×
136	**SI** T = long **Y** A = mt(many) **Y** Q = lt(medium) **ENTONCES** E = long				R4		✓
137	**SI** T = long **Y** A = mt(many) **Y** Q = lt(long) **ENTONCES** E = long						×

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#	Regla	7 s	17 s	27 s	37 s	47 s	Usada
138	**SI** T = long **Y** A = mt(many) **Y** Q = lt(very long) **ENTONCES** E = long						×
139	**SI** T = long **Y** A = mt(too many) **Y** Q = lt(very small) **ENTONCES** E = long						×
140	**SI** T = long **Y** A = mt(too many) **Y** Q = lt(small) **ENTONCES** E = long						×
141	**SI** T = long **Y** A = mt(too many) **Y** Q = lt(small plus) **ENTONCES** E = long						×
142	**SI** T = long **Y** A = mt(too many) **Y** Q = lt(medium) **ENTONCES** E = long						×
143	**SI** T = long **Y** A = mt(too many) **Y** Q = lt(long) **ENTONCES** E = long						×
144	**SI** T = long **Y** A = mt(too many) **Y** Q = lt(very long) **ENTONCES** E = long					R4	✓
145	**SI** T = very long **Y** A = mt(none) **Y** Q = lt(very small) **ENTONCES** E = very long						×
146	**SI** T = very long **Y** A = mt(none) **Y** Q = lt(small) **ENTONCES** E = very long						×
147	**SI** T = very long **Y** A = mt(none) **Y** Q = lt(small plus) **ENTONCES** E = very long						×
148	**SI** T = very long **Y** A = mt(none) **Y** Q = lt(medium) **ENTONCES** E = very long						×
149	**SI** T = very long **Y** A = mt(none) **Y** Q = lt(long) **ENTONCES** E = very long						×
150	**SI** T = very long **Y** A = mt(none) **Y** Q = lt(very long) **ENTONCES** E = very long						×
151	**SI** T = very long **Y** A = mt(a few) **Y** Q = lt(very small) **ENTONCES** E = very long						×
152	**SI** T = very long **Y** A = mt(a few) **Y** Q = lt(small) **ENTONCES** E = very long						×
153	**SI** T = very long **Y** A = mt(a few) **Y** Q = lt(small plus) **ENTONCES** E = very long						×
154	**SI** T = very long **Y** A = mt(a few) **Y** Q = lt(medium) **ENTONCES** E = very long						×
155	**SI** T = very long **Y** A = mt(a few) **Y** Q = lt(long) **ENTONCES** E = very long						×
156	**SI** T = very long **Y** A = mt(a few) **Y** Q = lt(very long) **ENTONCES** E = very long						×
157	**SI** T = very long **Y** A = mt(few) **Y** Q = lt(very small) **ENTONCES** E = very long						×
158	**SI** T = very long **Y** A = mt(few) **Y** Q = lt(small) **ENTONCES** E = very long						×
159	**SI** T = very long **Y** A = mt(few) **Y** Q = lt(small plus) **ENTONCES** E = very long						×
160	**SI** T = very long **Y** A = mt(few) **Y** Q = lt(medium) **ENTONCES** E = very long						×
161	**SI** T = very long **Y** A = mt(few) **Y** Q = lt(long) **ENTONCES** E = very long						×
162	**SI** T = very long **Y** A = mt(few) **Y** Q = lt(very long) **ENTONCES** E = very long						×
163	**SI** T = very long **Y** A = mt(medium) **Y** Q = lt(very small) **ENTONCES** E = very long						×
164	**SI** T = very long **Y** A = mt(medium) **Y** Q = lt(small) **ENTONCES** E = very long						×
165	**SI** T = very long **Y** A = mt(medium) **Y** Q = lt(small plus) **ENTONCES** E = very long						×
166	**SI** T = very long **Y** A = mt(medium) **Y** Q = lt(medium) **ENTONCES** E = very long						×
167	**SI** T = very long **Y** A = mt(medium) **Y** Q = lt(long) **ENTONCES** E = very long						×
168	**SI** T = very long **Y** A = mt(medium) **Y** Q = lt(very long) **ENTONCES** E = very long						×
169	**SI** T = very long **Y** A = mt(many) **Y** Q = lt(very small) **ENTONCES** E = very long	R5					✓
170	**SI** T = very long **Y** A = mt(many) **Y** Q = lt(small) **ENTONCES** E = very long		R5	R5			✓
171	**SI** T = very long **Y** A = mt(many) **Y** Q = lt(small plus) **ENTONCES** E = very long						×
172	**SI** T = very long **Y** A = mt(many) **Y** Q = lt(medium) **ENTONCES** E = very long						×
173	**SI** T = very long **Y** A = mt(many) **Y** Q = lt(long) **ENTONCES** E = very long						×
174	**SI** T = very long **Y** A = mt(many) **Y** Q = lt(very long) **ENTONCES** E = very long						×
175	**SI** T = very long **Y** A = mt(too many) **Y** Q = lt(very small) **ENTONCES** E = very long						×
176	**SI** T = very long **Y** A = mt(too many) **Y** Q = lt(small) **ENTONCES** E = very long						×
177	**SI** T = very long **Y** A = mt(too many) **Y** Q = lt(small plus) **ENTONCES** E = very long						×
178	**SI** T = very long **Y** A = mt(too many) **Y** Q = lt(medium) **ENTONCES** E = very long						×
179	**SI** T = very long **Y** A = mt(too many) **Y** Q = lt(long) **ENTONCES** E = very long				R5		✓
180	**SI** T = very long **Y** A = mt(too many) **Y** Q = lt(very long) **ENTONCES** E = very long					R5	✓

Fuente: etiquetas de T, A y Q de las Tablas II a IV del artículo original (p. 711), combinadas por la regla del producto ($5 \times 6 \times 6 = 180$); reglas resaltadas: Apéndice del artículo original (pp. 715 y 716). Código: `gen_tabla_180.py`; mismos datos en `resultados/reglas_180.csv`.

2. Los seis apartados del simulador

Las figuras siguientes muestran, ampliados, los seis apartados de la ventana del simulador (Fig. 4 del artículo principal). Corresponden al escenario 720/2160 veh/h, lectura calibrada y semilla 1, en $t = 118$ s (un instante posterior en 1 s al de la Fig. 4 del artículo principal), cuando el controlador decide extender 4 s el verde del brazo E-W en la intervención de 47 s, por la regla R2.

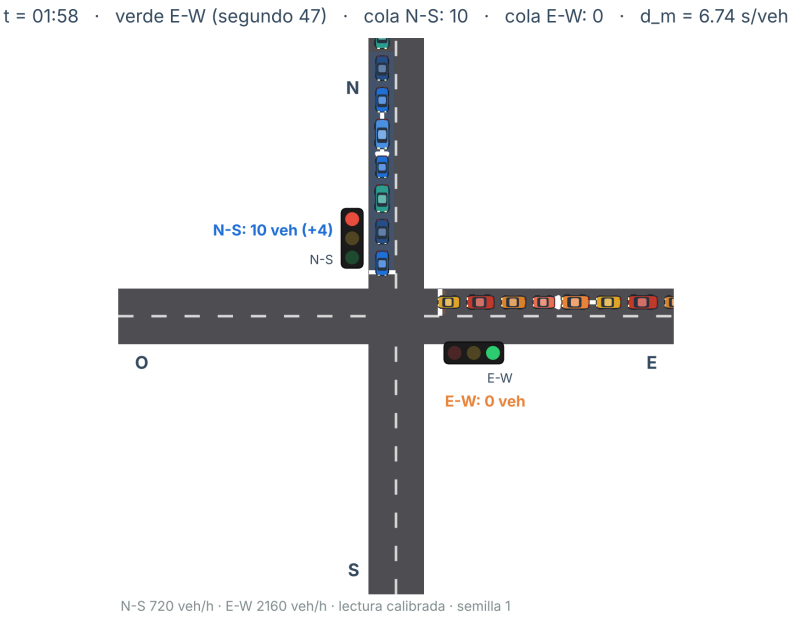


Figura S1: (a) La calle: cruce con vehículos en movimiento (azul: N-S; naranja: E-W), semáforos de tres luces, colas y encabezado con el tiempo y d_m.



Figura S2: (b) Panel de reglas de la intervención de 47 s: entradas de la decisión, activación de cada regla y regla que dispara (⋆, R2).

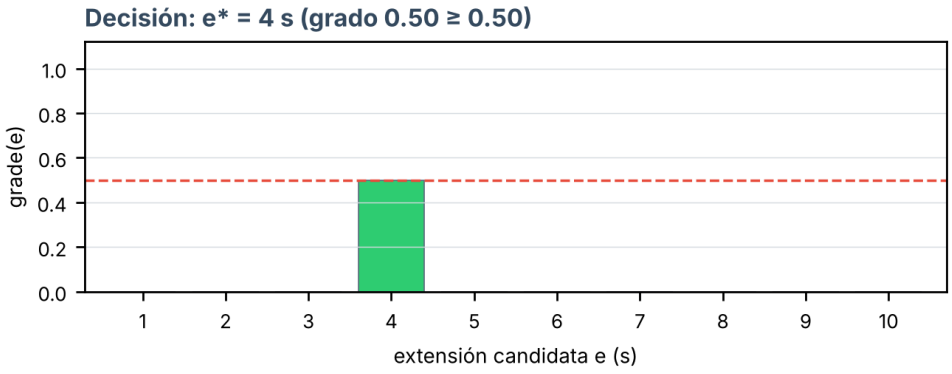


Figura S3: (c) Vector de grados $\gamma(e)$ de las extensiones candidatas, umbral 0.5 y extensión elegida ($e^* = 4\,\mathrm{s}$, grado 0.50).

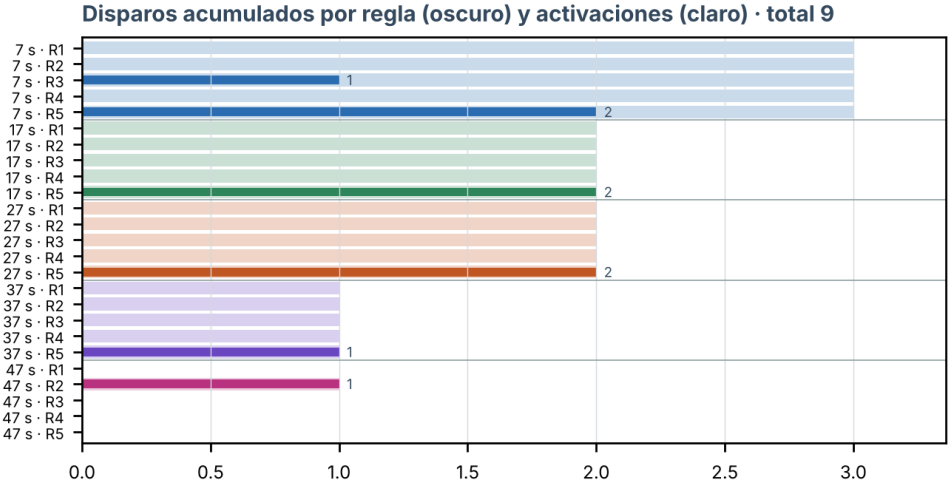


Figura S4: (d) Disparos acumulados (barra oscura) y activaciones (barra clara) de las 25 reglas, agrupadas por intervención, hasta $t = 118\,\mathrm{s}$.

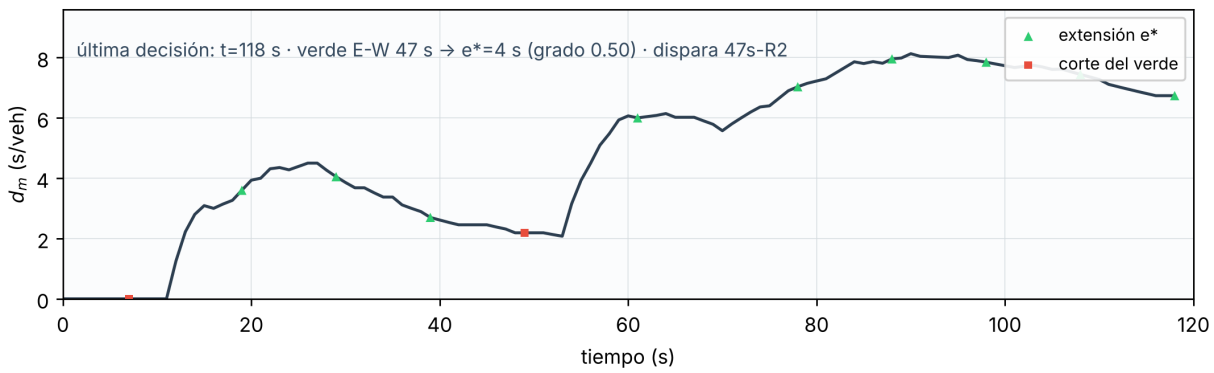


Figura S5: (e) Traza del retardo medio d_m en el tiempo. Los triángulos verdes marcan las extensiones del verde y los cuadrados rojos, los cortes.

Decisiones y regla que dispara

```
t=   7 N-S v 7s > CORTE (g=0.00)
t=  19 E-W v 7s > e*=10 [7s R5]
t=  29 E-W v17s > e*=10 [17s R5]
t=  39 E-W v27s > e*=10 [27s R5]
t=  49 E-W v37s > CORTE (g=0.00)
t=  61 N-S v 7s > e*= 5 [7s R3]
t=  78 E-W v 7s > e*=10 [7s R5]
t=  88 E-W v17s > e*=10 [17s R5]
t=  98 E-W v27s > e*=10 [27s R5]
t= 108 E-W v37s > e*=10 [37s R5]
t= 118 E-W v47s > e*= 4 [47s R2]
```

Figura S6: (f) Registro de decisiones: tiempo, brazo en verde, segundo de verde, extensión elegida y regla que dispara.

3. Figuras adicionales

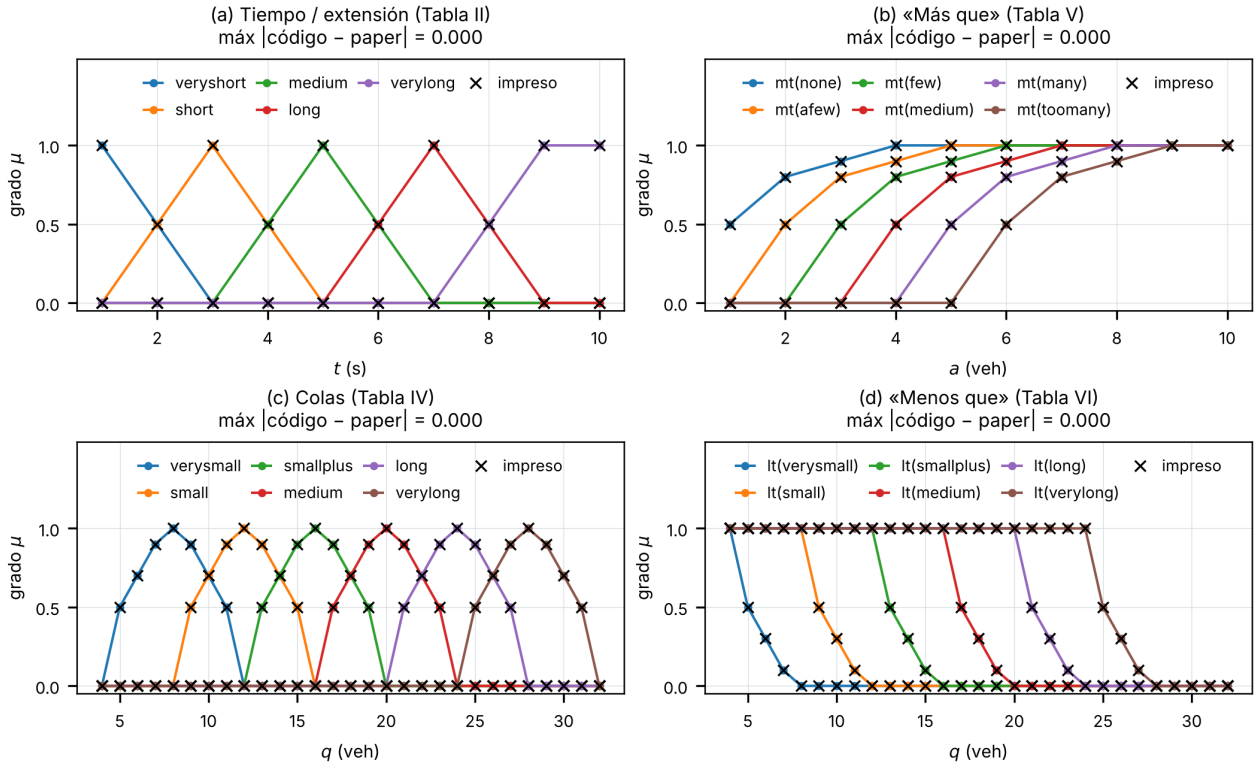


Figura S7: Validación de la fuzzificación: curvas del código (líneas) y valores impresos en el artículo original (×) para las Tablas II, IV, V y VI del original. La diferencia máxima es 0.000 en los cuatro paneles.

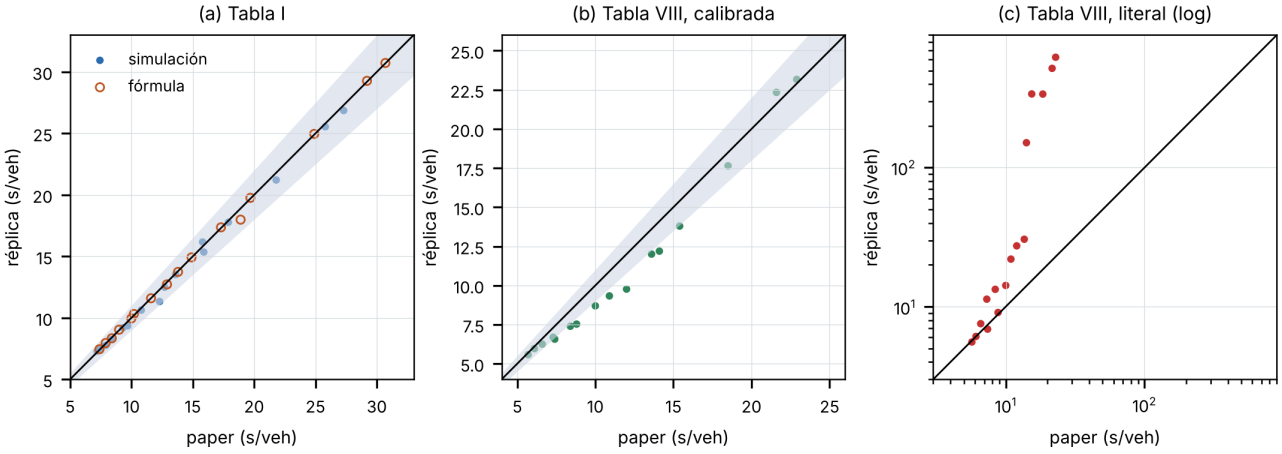


Figura S8: Réplica frente al artículo original (la diagonal es la coincidencia exacta; la banda es ±10 %): (a) Tabla I del original; (b) Tabla VIII del original con la lectura calibrada; (c) Tabla VIII del original con la lectura literal, en escala logarítmica.

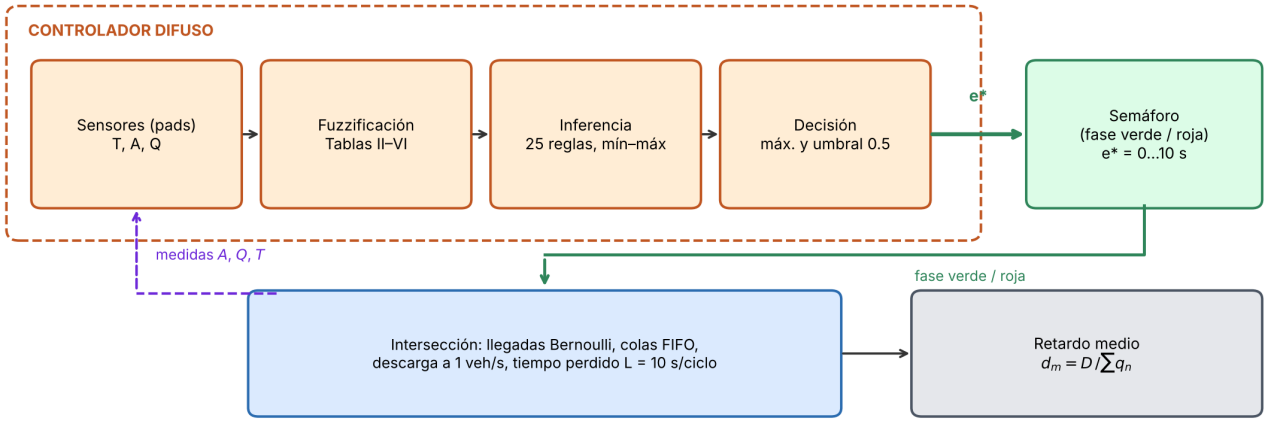


Figura S9: Diagrama de bloques del sistema de control de Pappis y Mamdani (elaboración propia a partir del artículo original).

4. Tablas adicionales

Tabla S2: Verificación numérica de las deducciones de las Secciones III y IV: cada fórmula se compara con un cálculo independiente.

Fórmula	Cálculo independiente	Resultado
(2), (4): $D = \sum_n$ colas	suma de esperas individuales (FIFO), 1000 escenarios	dif. máx. 0
Simulador: $\sum_n$ colas	espera de los que salieron + residual (8 corridas)	dif. máx. 0
Primer término de (7)	área de la cola de un modelo fluido	error 0.013 %
Segundo término de (7)	cola M/D/1 simulada (Lindley)	error 0.9 % (ruido)
Reparto g_i de (6)	igualdad $X_1 = X_2$	dif. $< 10^{-15}$
Ciclo C_0 de (6)	mínimo numérico de Webster	retardo ≤ 4.0 % sobre el mínimo
(8), (9): máx $= 1 - \mu_A$, mín $= 0$	234 puntos de 12 conjuntos	0 fallos
Regla composicional exacta	200 relaciones aleatorias	dif. máx. 0
Conteos	$5 \cdot 6 \cdot 6$; $5 \cdot 5$; $7 + 5 \cdot 10$	180; 25; 57

Fuente: `verificar_formulas.py` (semilla 2026); caudales de la Tabla VIII para la comparación de C_0.

Tabla S3: Efecto de borde del horizonte: aumento relativo del retardo medio al sumar la espera residual de los vehículos que siguen en cola en $t = H$ (20 réplicas de 7200 s, 16 flujos).

Controlador	Media (%)	Máximo (%)
Ciclo fijo	0.28	0.87
Difuso, lectura calibrada	0.17	0.33
Difuso, lectura literal (saturada)	8.1	51.8

Fuente: simulaciones propias (`efecto_borde.py`) con las mismas llegadas, con y sin `Sim.residual_delay`.